

Bank Customer Churn Analysis

Final Project Report

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Course: **TechMaster Certificate Program in Data Analytics**

- Skillup.online

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Executive Summary

This report presents an in-depth analysis of customer churn patterns for a retail bank using Power BI and Python.

The project aims to identify key factors influencing customer attrition and develop actionable insights to improve retention strategies. The dataset includes 10,000 customer records containing demographic, financial, and behavioral variables, analyzed to uncover trends across geography, gender, age, and activity levels.

Project Objective & Dataset Overview

The primary objective of this project is to analyze bank customer churn and understand which factors contribute most to attrition. The dataset consists of 10,000 records with features like Credit Score, Geography, Gender, Age, Balance, Tenure, Number of Products, and Churn Status. Data cleaning, exploratory analysis, and visual insights were generated to build a clear understanding of customer behavior patterns.

Tools & Methodology

Tools Used:

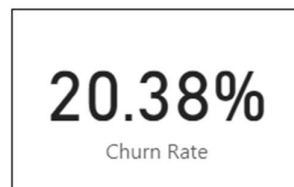
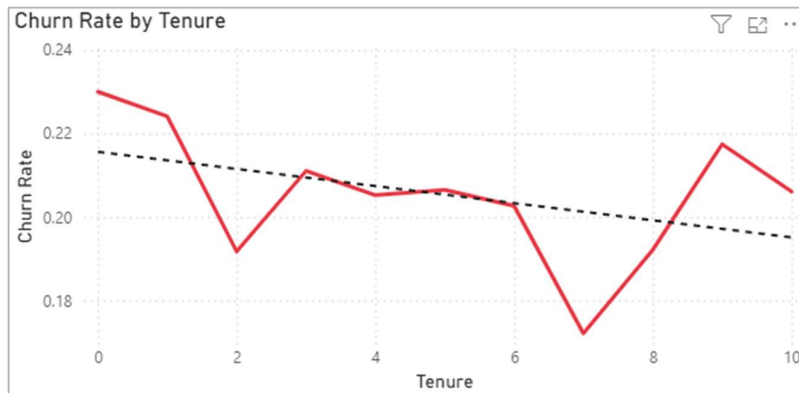
- Power BI - For creating dynamic dashboards and KPI visualizations.
- Python (Pandas, Matplotlib, Seaborn) - For data cleaning, statistical analysis, and visual exploration.

Methodology:

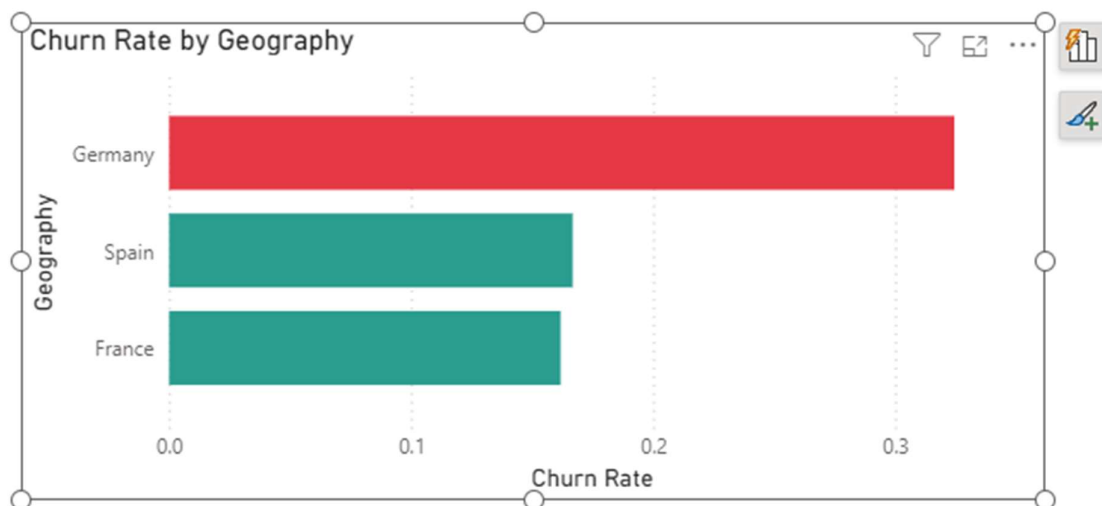
1. Data Cleaning - Checked for null values, duplicates, and inconsistencies.
2. Exploratory Data Analysis - Identified churn patterns across demographic and financial attributes.
3. Dashboard Development - Designed interactive Power BI dashboards showing KPIs like churn rate, active user ratio, and average balance.

Section A: Power BI – Business Intelligence & Dashboards

1. What is the overall churn rate, and how does it trend over time (if applicable)?



2. How does churn vary by geography (France, Germany, Spain)?



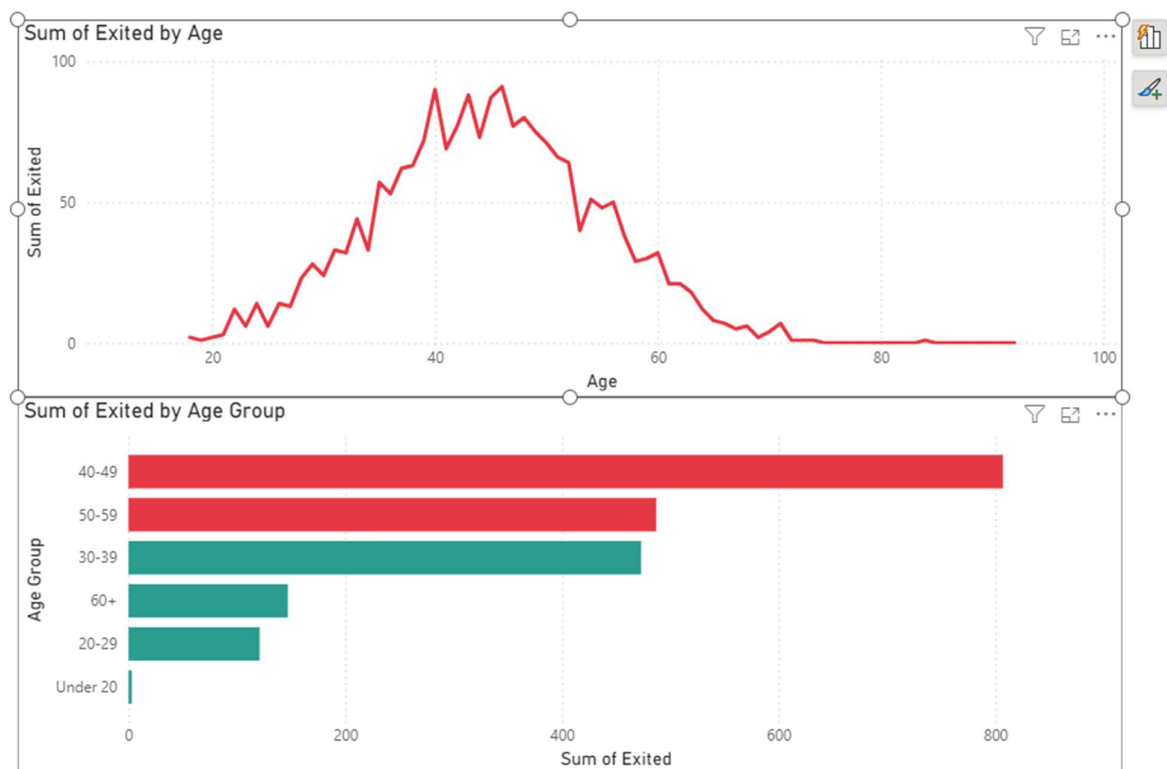
the churn rate by geography from your dataset: **France = 16.15%, Germany = 32.44%, Spain= 16.67%**

3. What are the differences in churn rates between male and female customers?



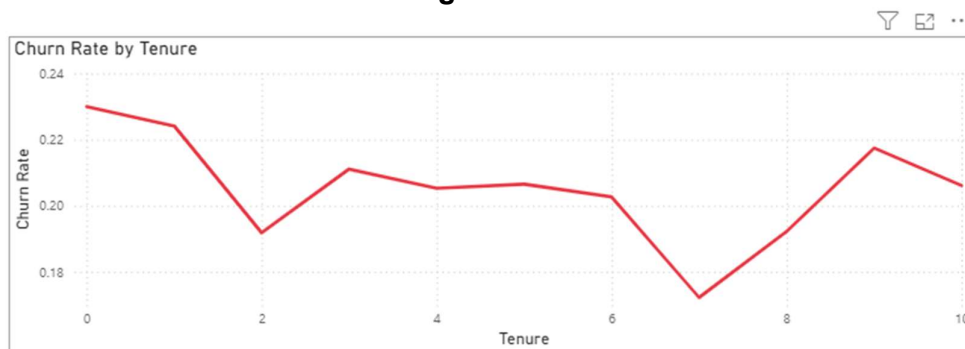
Churn Rate is Higher in Female (1139) then as compare to Male (899)

4. What is the churn rate distribution across different age groups?

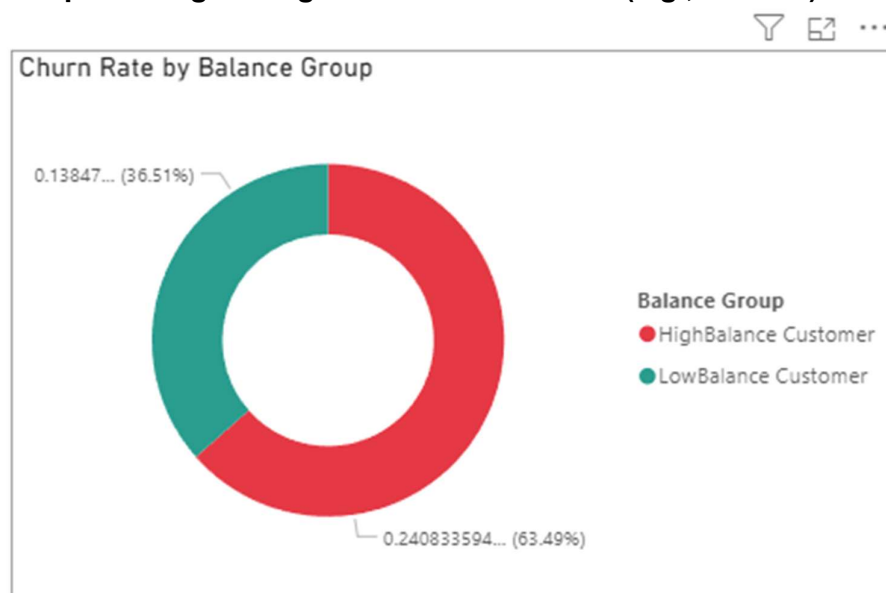


Age group from 40-49 has highest churn rate as per data

5. How does churn behavior change with tenure?

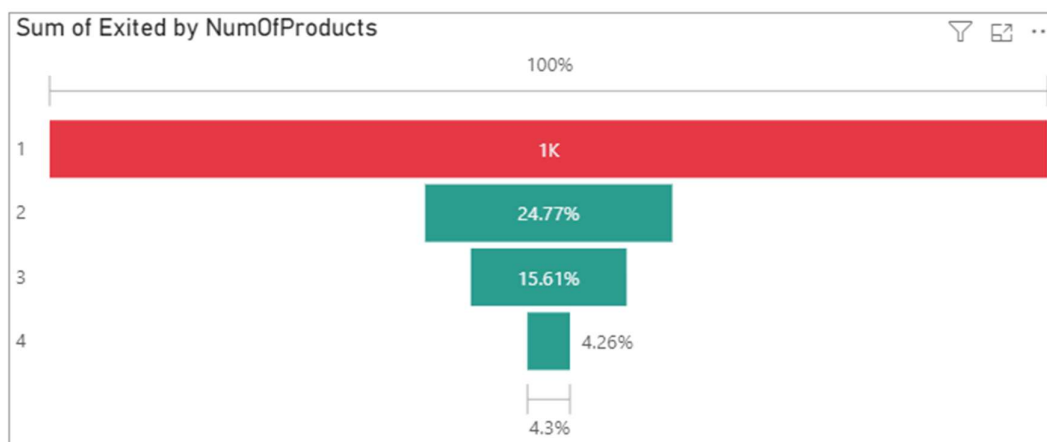


6. What percentage of high-balance customers (e.g., >\$100K) have churned?



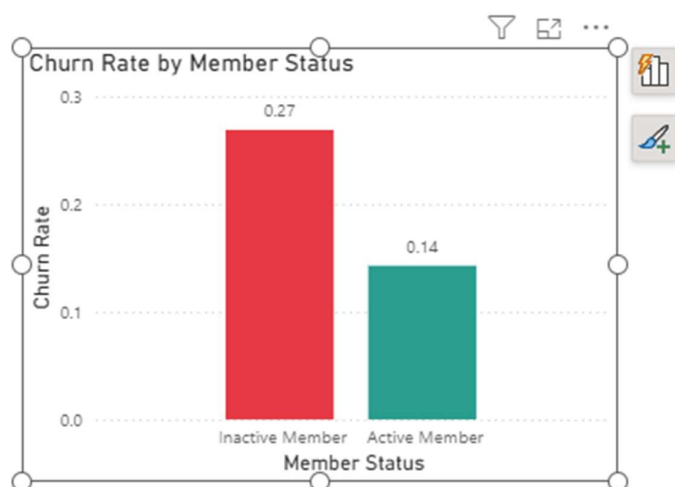
36.51% of High-balance Customers have churned.

7. How do churn rates compare across different NumOfProducts?

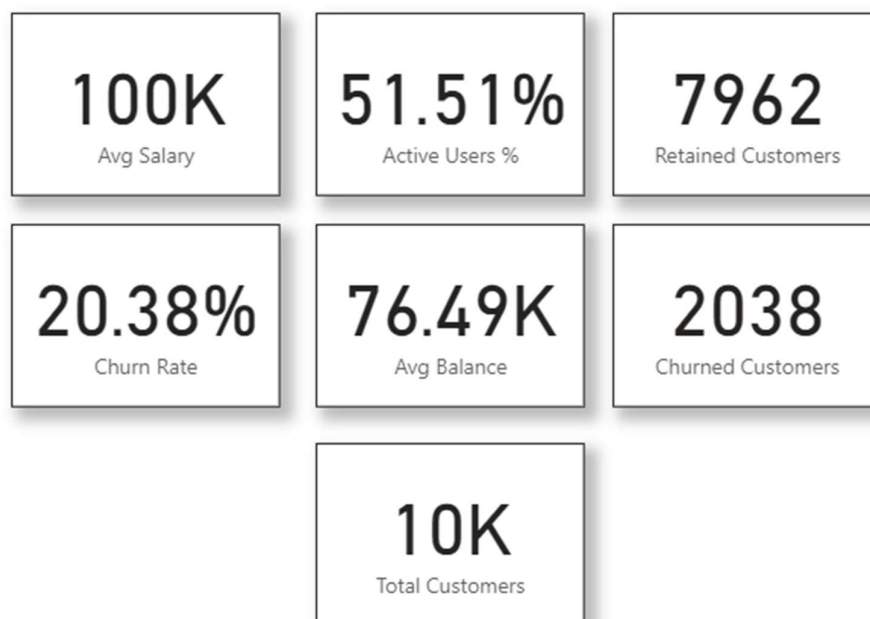


Customers using only one product show a significantly higher likelihood of churn.

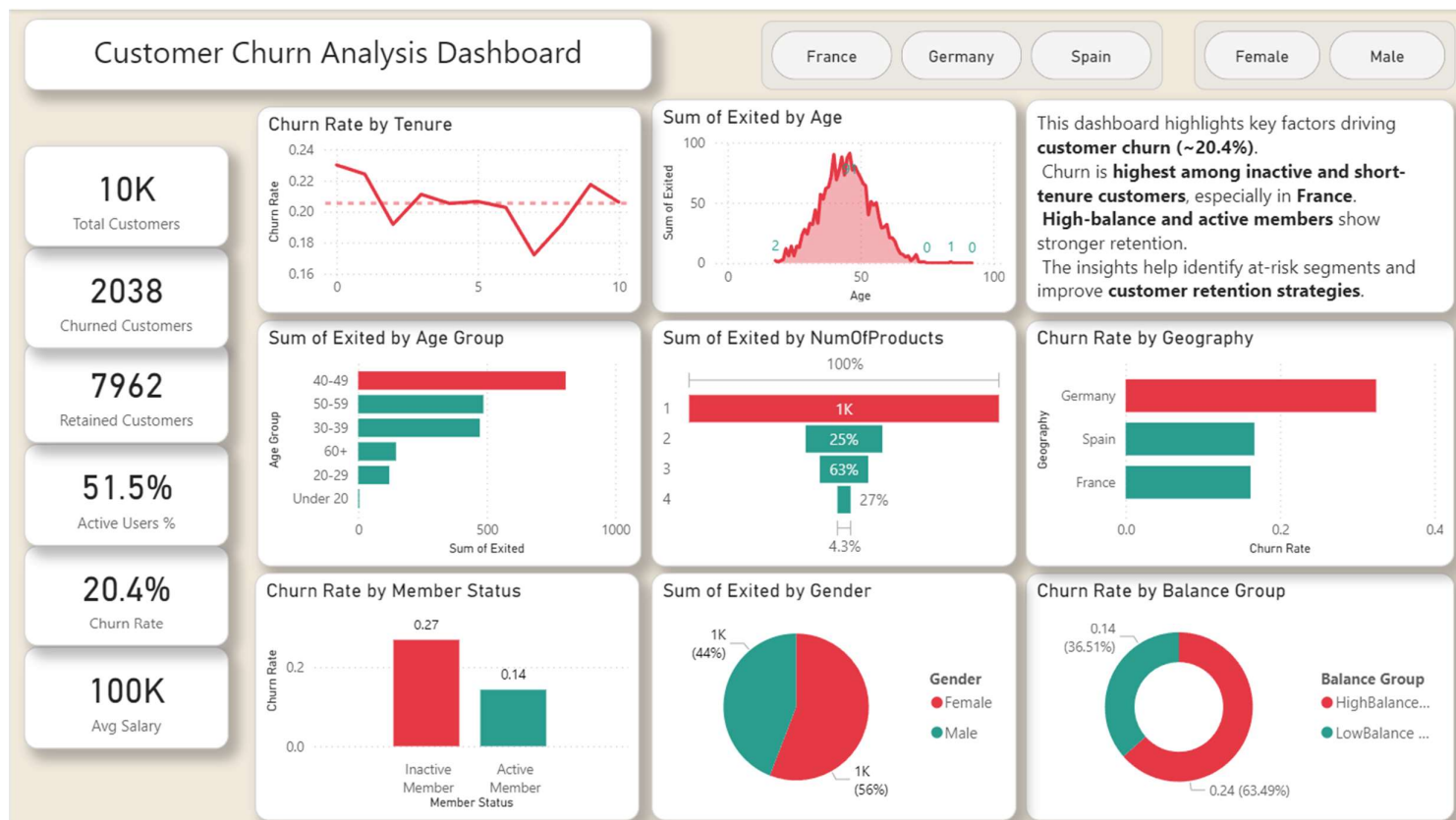
8. Are inactive members (IsActiveMember = 0) more likely to churn?



9. Can we build a KPI dashboard to track churn metrics like average salary, balance, and active users?

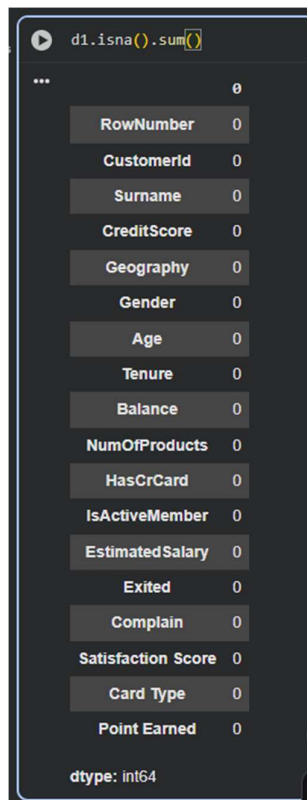


10. What does a customer persona dashboard (churned vs. retained) look like?



Section B: Python – Exploratory Data Analysis (Provide a description of your answers in all the cells explaining the outputs)

11. Check the percentage of missing data and handle accordingly.



```
d1.isna().sum()
```

...	0
RowNumber	0
CustomerId	0
Surname	0
CreditScore	0
Geography	0
Gender	0
Age	0
Tenure	0
Balance	0
NumOfProducts	0
HasCrCard	0
IsActiveMember	0
EstimatedSalary	0
Exited	0
Complain	0
Satisfaction Score	0
Card Type	0
Point Earned	0

dtype: int64

no missing data.

12. How many rows and columns are there in the dataset?



```
print("Rows:", df.shape[0])  
print("Columns:", df.shape[1])
```

Rows: 10000
Columns: 18

13. What is the distribution of churned vs. non-churned customers?

```
[20] count_dis = df["Exited"].value_counts()

[21] count_dis.index = ["Non-Churned", "Churned"]

[22] percent_distribution = (count_dis / len(df)) * 100

print("Count Distribution:\n", count_dis)
print("\nPercentage Distribution:\n", percent_distribution.round(2))
```

... Count Distribution:
Non-Churned 7962
Churned 2038
Name: count, dtype: int64

Percentage Distribution:
Non-Churned 79.62
Churned 20.38
Name: count, dtype: float64

14. What is the distribution of EstimatedSalary of churned and retained Customers?

```
salary_distribution = df.groupby("Exited")["EstimatedSalary"].describe()

print(salary_distribution)
```

	count	mean	std	min	25%	50%	\
Exited							
0	7962.0	99726.853141	57399.956717	90.07	50783.295	99620.355	
1	2038.0	101509.908783	57932.623392	11.58	51924.020	102489.335	
		75%	max				
Exited							
0	148602.4450	199992.48					
1	152443.8575	199808.10					

15. How do churn rates vary by Gender, Geography, and IsActiveMember?

```
# Function to calculate churn rate
def churn_rate(group):
    return (group["Exited"].mean() * 100).round(2)

# Churn by Gender
churn_by_gender = df.groupby("Gender").apply(churn_rate)
print("Churn Rate by Gender:\n", churn_by_gender, "\n")

# Churn by Geography
churn_by_geo = df.groupby("Geography").apply(churn_rate)
print("Churn Rate by Geography:\n", churn_by_geo, "\n")

# Churn by Active/Inactive Members
churn_by_active = df.groupby("IsActiveMember").apply(churn_rate)
print("Churn Rate by Active/Inactive Members:\n", churn_by_active)
```

... Churn Rate by Gender:

Gender	Churn Rate
Female	25.07
Male	16.47

dtype: float64

Churn Rate by Geography:

Geography	Churn Rate
France	16.17
Germany	32.44
Spain	16.67

dtype: float64

Churn Rate by Active/Inactive Members:

IsActiveMember	Churn Rate
0	26.87
1	14.27

dtype: float64

16. What is the average CreditScore, Balance, and EstimatedSalary of churned vs. retained customers?

```
avg_values = df.groupby("Exited")[["CreditScore", "Balance", "EstimatedSalary"]].mean().round(2)

avg_values.index = ["Non-Churned", "Churned"]
```

avg_values

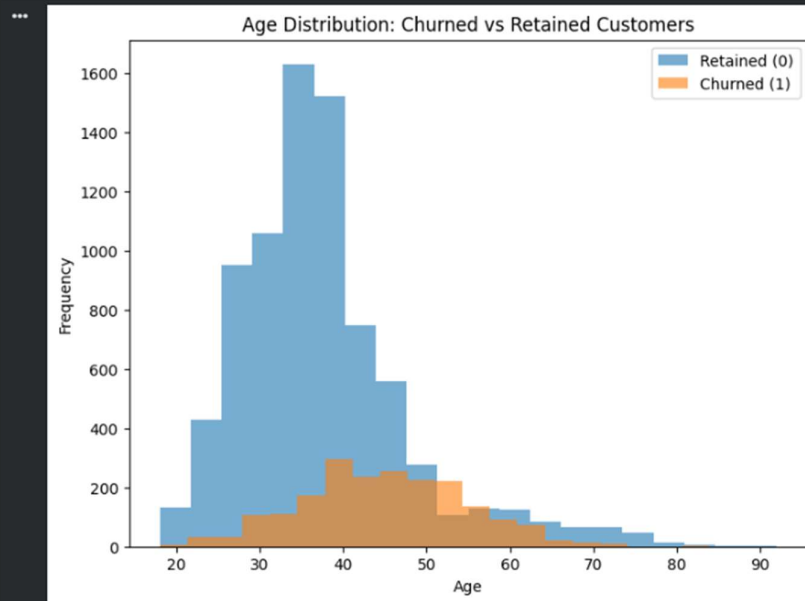
	CreditScore	Balance	EstimatedSalary
Non-Churned	651.84	72742.75	99726.85
Churned	645.41	91109.48	101509.91

Next steps: [Generate code with avg_values](#) [New interactive sheet](#)

17. How does Age impact churn? Plot histograms and boxplots for churned and non-churned groups.

```
Churned = df[df["Exited"] == 1] ["Age"]
Retained = df[df["Exited"] == 0] ["Age"]
```

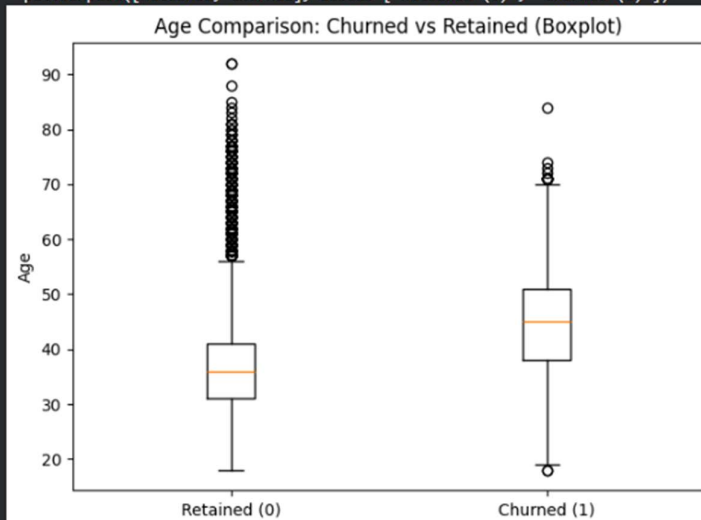
```
plt.figure(figsize=(8, 6))
plt.hist(Retained, bins=20, alpha=0.6, label="Retained (0)")
plt.hist(Churned, bins=20, alpha=0.6, label="Churned (1)")
plt.title("Age Distribution: Churned vs Retained Customers")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.legend()
plt.show()
```



```
plt.figure(figsize=(7, 5))
plt.boxplot([Retained, Churned], labels=["Retained (0)", "Churned (1)"])

plt.title("Age Comparison: Churned vs Retained (Boxplot)")
plt.ylabel("Age")
plt.show()
```

*** /tmp/ipython-input-3827525609.py:2: MatplotlibDeprecationWarning: The 'labels' parameter
plt.boxplot([Retained, Churned], labels=["Retained (0)", "Churned (1)"])



18. Is there any correlation among numeric features like CreditScore, Balance, and EstimatedSalary?

```
[25] numeric_df = df[['CreditScore', 'Balance', 'EstimatedSalary']]
[26] corr_matrix = numeric_df.corr()
[27] corr_matrix
```

	CreditScore	Balance	EstimatedSalary
CreditScore	1.000000	0.006268	-0.001384
Balance	0.006268	1.000000	0.012797
EstimatedSalary	-0.001384	0.012797	1.000000

Next steps: [Generate code with corr_matrix](#) [New interactive sheet](#)

22. Are customers with only one product (NumOfProducts = 1) more likely to churn than those with multiple?



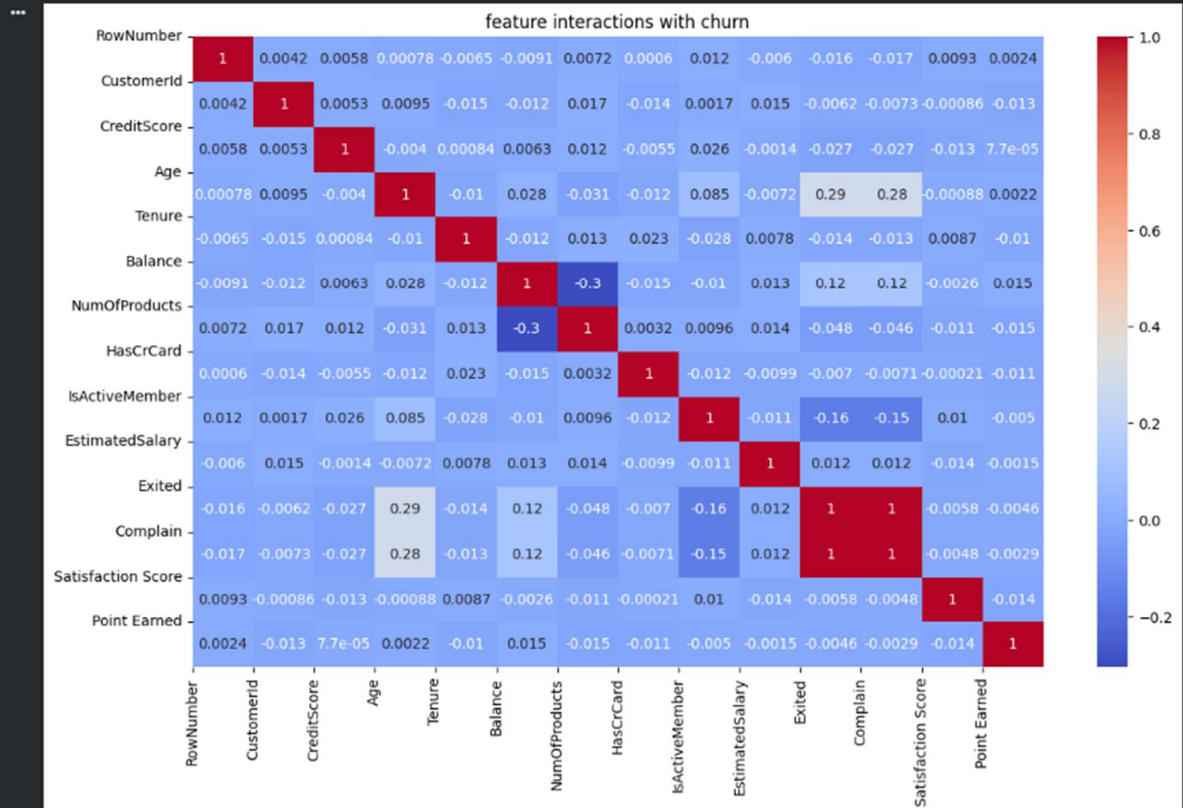
- If **NumOfProducts = 1** has the highest churn rate → customers with only one product are more likely to churn.
- If **multiple-product customers** (2, 3, 4) have lower churn → they stay longer and are more loyal.

19. What does a heatmap reveal about feature interactions with churn?

```
corr = num_df.corr()
plt.figure(figsize=(12, 8))

plt.title("feature interactions with churn")
sns.heatmap(corr, annot=True, cmap='coolwarm')
ticks = np.arange(len(corr.columns))
plt.xticks(ticks, corr.columns, rotation=90)
plt.yticks(ticks, corr.columns)

plt.tight_layout()
plt.show()
```



Key Insights & KPI Highlights

- Overall churn rate: Approximately 20.34% of total customers.
- Geography: Germany showed the highest churn rate, followed by France and Spain (Figure 1).
- Gender: Female customers exhibited slightly higher churn rates compared to males (Figure 2).
- Age: Higher churn observed among customers aged 40-60 (Figure 3).
- Tenure: Customers with shorter tenure (<5 years) were more likely to churn (Figure 4).
- High-Balance Customers: Despite higher balances, customers with >\$100K balance showed on table churn.
- Products: Customers with only one product had the highest churn rate (Figure 5).
- Inactive Members: Non-active customers (IsActiveMember = 0) were more likely to leave (Figure 6).

Customer Persona & Churn Patterns

Based on Power BI segmentation, the typical churned customer profile includes:

- Middle-aged (40-55 years)
- Moderate CreditScore (~610-670)
- High Balance (> \$100,000)
- 1 Product, Non-Active Member, Usually from Germany

These insights help the bank create retention-focused campaigns targeting this high-risk segment.

Conclusion & Recommendations

The analysis provides a data-driven understanding of customer churn dynamics. The following steps are recommended:

1. Launch retention programs for customers in Germany with high balances and low engagement.
2. Introduce multi-product incentives to reduce single-product churn.
3. Target middle-aged customers with personalized offers.
4. Improve engagement campaigns for inactive members through digital touchpoints.